



Assessment task 4 - Professional practice and contribution

Creating a Real-Time Digital Twin of a Truss Bridge

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1. Project Overview

This journal documents my professional engagement and technical contributions during Capstone Project A: Real-Time Digital Twin of a Truss Bridge.

The project's objective is to design a MATLAB-based digital twin that replicates the behaviour of a laboratory truss bridge in real time by integrating sensor data — strain, acceleration, and displacement — with a physics-based structural model. The system is intended to enable structural health monitoring (SHM) by providing continuous, data-driven insights into bridge performance.

Our team of five — Atal Azadzoi, Rouhaifa Kakhie, Shahzad Ahmed, Shivam Raj, and myself — worked under the supervision of Professor Ricky Chan. My responsibilities consisted of contributing to the design and development of the system framework, focusing on how the data pipeline, software, and sensing components could be integrated into a digital twin. Alongside this, I also played a key role in the research phase, reviewing literature on SHM and exploring methods for combining sensor data with physics-based models.

Capstone A (Weeks 1–13) concentrated on research, conceptual modelling, and system design. Capstone B (Weeks 14–26) will build upon this foundation through implementation, sensor integration, and experimental validation.

2. Key Figures



Figure 1. Prototype robust bridge (photograph provided by Professor Ricky Chan)

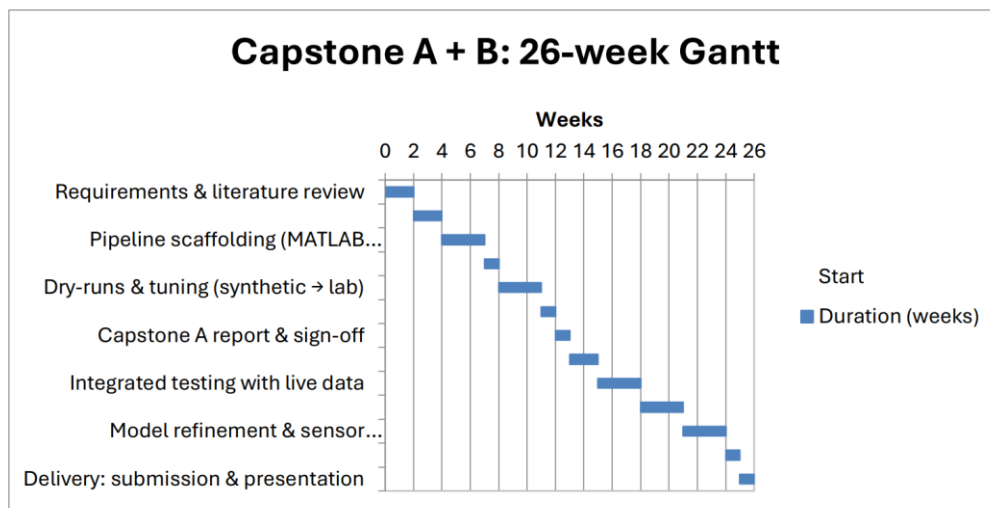


Figure 2. Official 26-week Gantt Chart (Project Schedule)

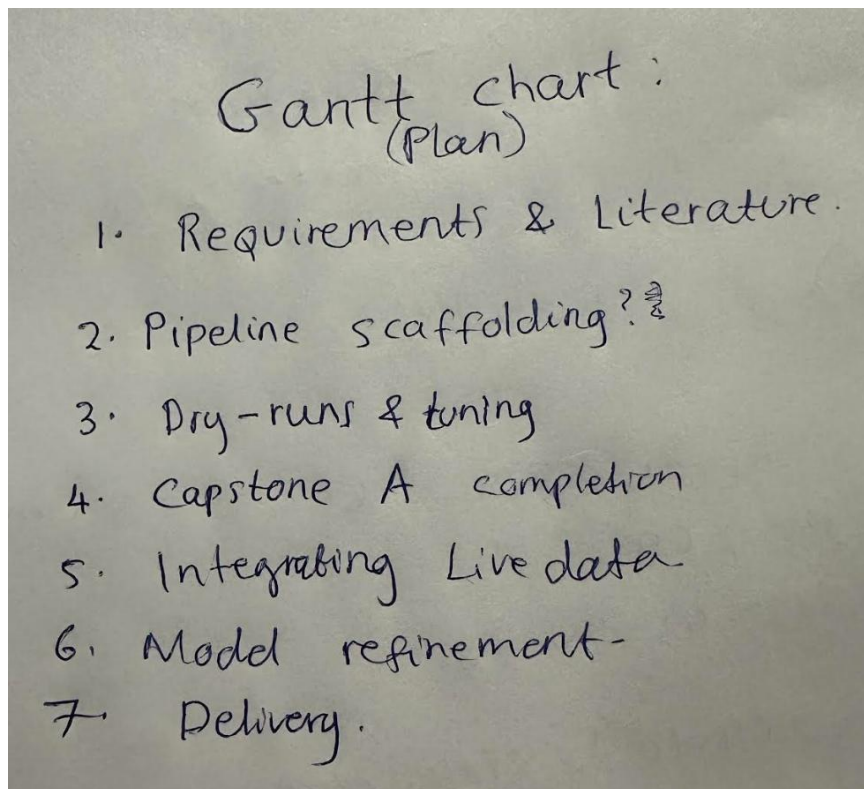


Figure 3. Gantt Chart – Initial hand-drawn planning sketch

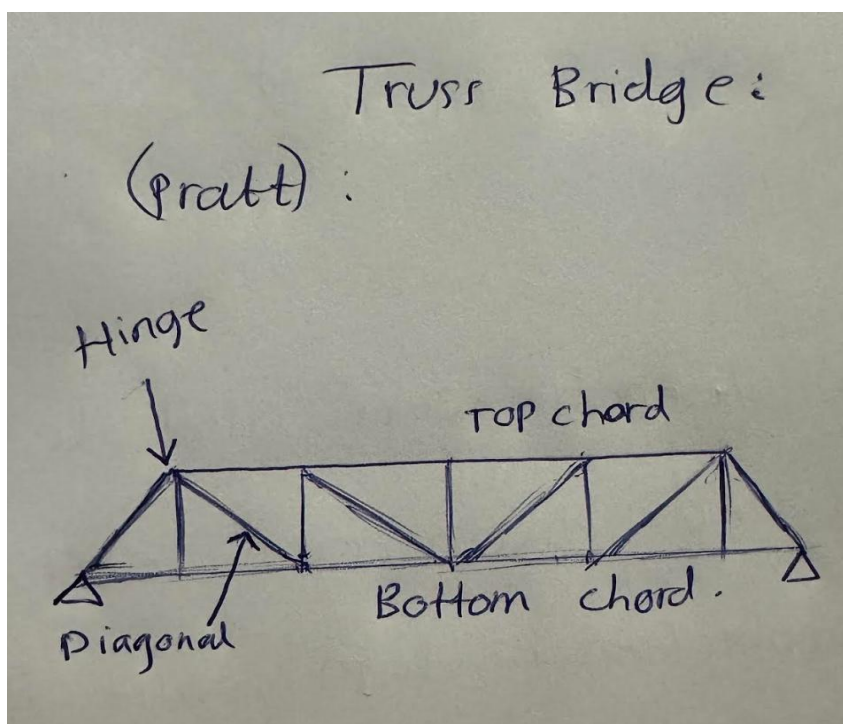


Figure 4. Generic hand-drawn Truss Bridge concept sketch

Matlab Structure Array Concept

Bridge

Node number
sensor type
Data [strain, Displacement, LVDT]
simulated data
units ??
simulated time.

Figure 5. MATLAB structure array example for digital-twin data pipeline

3. Weekly Summary of Work and Reflection

Weeks 3 | Problem Definition and Research Foundation

Fortnight 1–2

During the first fortnight, our team worked with the supervisor to clarify the project objectives and establish four guiding research questions. My contribution focused on investigating the feasibility of implementing real-time data acquisition methods in MATLAB. I also conducted multiple literature reviews on digital twins, structural health monitoring, and sensor placement strategies, and summarised the findings in our shared team document.

Reflection:

These early meetings highlighted the importance of balancing curiosity with practicality. By clearly understanding the project's constraints, I was able to help shape realistic goals and ensure the scope remained achievable, avoiding unnecessary over-engineering.

Weeks 3–4 | Design Concept and Team Coordination

Following feedback from Professor Chan, we refined our research questions and formally divided responsibilities within the team. My contribution was assisting the team draft a conceptual MATLAB workflow diagram to illustrate how sensor data would progress through acquisition, quality-control, and estimation modules. Aswell, the team produced a Gantt chart to map out key milestones across both Capstone A and B. In addition I also did further research on the previous articles we collected.

Reflection:

Developing the software pipeline at this stage clarified the interdependencies between sensing, modelling, and computation. This process reinforced the importance of establishing a robust system before moving into detailed coding or implementation.

Weeks 5–6 | MATLAB Framework and Supervisor Feedback

In Weeks 5–6, I began assisting developing a prototype acquisition broker in MATLAB to simulate incoming strain and acceleration data streams. Using key information on articles which carried out similar tests. To support this, I designed a structured data-logging format using arrays to store both timestamps and sensor readings.

During this period, Professor Chan reviewed our draft report and highlighted that it lacked cohesion, as the initial sections had been written independently. In response, we restructured the document collaboratively and agreed on consistent terminology across all technical sections.

Reflection:

Working on both technical coding and group writing and reporting in parallel improved my awareness of how individual contributions must integrate into a coherent whole. The supervisor's constructive feedback helped guide not only the technical depth of our writing but also the clarity and professional tone required

Week 7 | Rebuild and Collaborative Editing

This week, our team rebuilt the report from the ground up, placing emphasis on academic polish, clarity, and logical flow. While other members also deepened the literature review, I focused on maintaining clarity across the document — managing cross-referencing, figure formatting, and ensuring consistency between technical sections. I continued assisting with the digital design framework in MATLAB and guided the integration of relevant research findings to strengthen the report's academic foundation. This week marked our transition from individually drafted sections to a truly collaborative team effort.

Reflection:

Shifting towards structured revision cycles highlighted the importance of clear communication and document control, practices that reflect professional engineering standards. I developed stronger coordination skills through shared editing on OneDrive and gained experience balancing technical design tasks with the responsibilities of leading research direction.

Week 8 | Research Expansion and MATLAB Learning

This week, I explored potential approaches for achieving real-time estimation within the digital twin. I trialled Kalman filtering and recursive least-squares algorithms, testing small MATLAB scripts with random-input data to observe how estimation residuals behaved. Alongside this, the team began drafting conceptual diagrams, including sensor-placement sketches and MATLAB data-flow charts. I contributed by linking these designs back to relevant research, ensuring the theoretical methods aligned with our system objectives and to previous research.

Reflection:

Bridging theoretical research with early-stage prototype coding strengthened my understanding of how SHM systems operate in practice. Experimenting directly in MATLAB proved more effective for learning than relying solely on literature review. This week also reinforced my role in guiding research direction while assisting in digital design, ensuring our system framework is sound.

Weeks 9–10 | Parallel Workstreams and Integration

To manage the workload more effectively, we divided into smaller groups. I worked alongside Rouhaifa and Shahzad on MATLAB model development, while Shivam and I also worked on polishing the documentation, risk planning, and visuals. Our sub-team built a basic finite-element (FE) truss model and tested how different stiffness and load values influenced bridge deflection. As lead researcher, I ensured our modelling approach aligned with the guiding research questions, while also contributing technically by adding validation and error-checking functions to improve the reliability of outputs.

Reflection:

This phase allowed me to strengthen both my coding and research skills. Pair-programming in MATLAB improved my problem-solving under time pressure, while linking our work back

to the research ensured the model development remained relevant to the project's broader goals

Week 11 | Final Integration and Submission Preparation

In the final week, the focus was on merging all report sections, polishing formatting, and checking compliance with submission requirements. My main responsibility was managing the report itself — coordinating version control, maintaining consistency in tone and formatting, and integrating the MATLAB development notes and diagrams into the methodology section. I also contributed through proofreading, ensuring the document met academic standards for clarity and presentation. As a team, we discussed potential directions for Capstone B, such as live sensor integration, latency measurement, and validating the model against physical tests.

Reflection:

Taking the lead on report coordination improved my skills in document management and academic writing. Integrating MATLAB content into the report gave me a clearer understanding of how technical work needs to be communicated, not just developed. This stage reinforced the importance of professional editing and version control to produce a cohesive final submission.

4. Project Minutes and Action Items

Meeting 1	
Project Name: Real-Time Digital Twin of a Truss Bridge Date of Meeting: 01 / 08 / 2025	
Location:	Microsoft Teams
Time:	12 : 00 pm – 12 : 45 pm
Minutes:	45 min
Attendance at Meeting	
Name	
Prof Ricky Chan	
Atal Azadzo	
Jarred Apps	
Rouhaifa Kakhie	
Shahzad Ahmed	
Shivam Raj	
Discussion	
Topic	Discussed
Project Introduction	• Supervisor outlined project expectations and assessment tasks. • Team confirmed the project scope: designing a MATLAB-based digital twin for a truss bridge prototype. • Discussed deliverables, structure, and Gantt chart planning.
Communication Plan	• Weekly team meetings scheduled; shared OneDrive folder created for collaboration. • Clarified individual strengths and potential role assignments.
Actioned Items	
Action	Owner / s
Begin literature review on digital twin applications and SHM systems.	All members
Draft preliminary problem statement and objectives.	Atal, Rouhaifa
Design first version of Gantt chart.	Jarred
Confirm team communication and file-sharing setup.	Shivam, Shahzad

Meeting 2			
Project Name:	Real-Time Digital Twin of a Truss Bridge	Date of Meeting:	08 / 08 / 2025
Location:	Microsoft Teams	Time:	12 : 00 pm – 12 : 45 pm
Minutes:	45 min		
Attendance at Meeting			
Name			
Prof Ricky Chan			
Atal Azadzoi			
Jarred Apps			
Rouhaifa Kakhie			
Shahzad Ahmed			
Shivam Raj			
Discussion			
Topic	Discussed		
Literature Progress	- Reviewed each member’s literature findings on SHM, data calibration, and MATLAB frameworks. - Supervisor advised citing recent journal sources and refining methodology justification.		
Task Planning	- Agreed on a detailed weekly timeline for Assessment 1. - Confirmed focus areas: MATLAB setup (Atal), literature writing (Rouhaifa), scheduling (Jarred), documentation (Shivam), and QA (Shahzad).		
Actioned Items			
Action	Owner / s		
Summarise 3 academic papers each for the shared literature table.	All members		
Write methodology draft connecting MATLAB workflow to literature.	Atal, Rouhaifa		
Update Gantt chart and submit to supervisor.	Jarred		
Compile citations in APA 7th style.	Shivam		

Meeting 3			
Project Name:	Real-Time Digital Twin of a Truss Bridge	Date of Meeting:	15 / 08 / 2025
Location:	Teams Call	Time:	11 : 30 am – 12 : 15 pm
Minutes:	45 min		
Attendance at Meeting			
Name			
Prof Ricky Chan			
Atal Azadzoi			
Jarred Apps			
Rouhaifa Kakhie			
Shahzad Ahmed			
Shivam Raj			
Discussion			
Topic	Discussed		
Supervisor Feedback	• Supervisor reviewed the first draft of Assessment 1 and provided feedback on structure and alignment with objectives. • Suggested linking literature insights with methodology flowcharts.		
Technical Framework	• Discussed MATLAB simulation structure and storage method using arrays. • Agreed to include flowcharts in the report illustrating the data acquisition process.		
Actioned Items			
Action	Owner / s		
Redraft methodology section and include new flowchart.	Shivam, Rouhaifa		
Refine introduction and problem definition.	Atal		
Prepare updated risk register and submit progress to supervisor.	Jarred, Shahzad		

Meeting 4			
Project Name:	Real-Time Digital Twin of a Truss Bridge	Date of Meeting:	22 / 08 / 2025
Location:	Microsoft Teams	Time:	1 : 00 pm – 1 : 45 pm
Minutes:	45 min		
Attendance at Meeting			
Name			
Prof Ricky Chan			
Atal Azadzoi			
Jarred Apps			
Rouhaifa Kakhie			
Shahzad Ahmed			
Shivam Raj			
Discussion			
Topic	Discussed		
Progress Review	• Supervisor reviewed report updates and approved technical direction. • Recommended unifying captions, formatting, and reference styles. • Discussed need for clearer explanation of MATLAB’s modular design.		
Code Development			
Actioned Items			
Action	Owner / s		
Improve MATLAB structure array for future scalability.	Atal		
Proofread and standardise formatting throughout the document.	Shivam, Rouhaifa		
Update timeline and add progress notes.	Jarred, Shahzad		

Meeting 5			
Project Name:	Real-Time Digital Twin of a Truss Bridge	Date of Meeting:	29 / 08 / 2025
Location:	Microsoft Teams	Time:	12 : 00 pm – 12 : 45 pm
Minutes:	45 min		
Attendance at Meeting			
Name			
Prof Ricky Chan			
Atal Azadzoi			
Jarred Apps			
Rouhaifa Kakhie			
Shahzad Ahmed			
Shivam Raj			
Discussion			
Topic	Discussed		
Documentation	• Integrated calibration and validation results into the draft report and ensured consistency across appendices.		
Actioned Items			
Action	Owner / s		
Implement calibration and validation section in report.	Shahzad, Rouhaifa		
Record calibration data plots for appendices.	Atal		
Update captions and ensure consistent labelling.	Shivam		
Review section for technical accuracy.	Jarred		

Meeting 6			
Project Name:	Real-Time Digital Twin of a Truss Bridge	Date of Meeting:	05 / 09 / 2025
Location:	Microsoft Teams	Time:	12 : 00 pm – 12 : 45 pm
Minutes:	45 min		
Attendance at Meeting			
Name			
Prof Ricky Chan			
Atal Azadzoi			
Jarred Apps			
Rouhaifa Kakhie			
Shahzad Ahmed			
Shivam Raj			
Discussion			
Topic	Discussed		
QA Implementation	• Team discussed QA section and reflection for journal entry.		
Progress Update	• Report structure 90% complete; only proofreading and referencing remain.		
Actioned Items			
Action	Owner / s		
Add timestamp and outlier detection in MATLAB script.	Atal		
Write QA discussion and results summary.	Shahzad		
Conduct full proofreading of the report.	Rouhaifa, Shivam		
Review final layout and figure alignment.	Jarred		

Meeting 7			
Project Name:	Real-Time Digital Twin of a Truss Bridge	Date of Meeting:	12 / 09 / 2025
Location:	Microsoft Teams	Time:	10 : 00 am – 10 : 45 am
Minutes:	45 min		
Attendance at Meeting			
Name			
Prof Ricky Chan			
Atal Azadzoi			
Jarred Apps			
Rouhaifa Kakhie			
Shahzad Ahmed			
Shivam Raj			
Discussion			
Topic	Discussed		
Mid-Semester Review	• Suggested adding annotations to graphs and code snippets.		
Next Steps	• Prepared reflections for Assessment 4 journal; began summarising individual contributions.		
Actioned Items			
Action	Owner / s		
Annotate MATLAB figures with explanations.	Atal		
Draft personal reflection for journal.	All members		
Finalise calibration data visualisations.	Jarred, Shahzad		

Meeting 8			
Project Name:	Real-Time Digital Twin of a Truss Bridge	Date of Meeting:	19 / 09 / 2025
Location:	Microsoft Teams	Time:	11 : 00 am – 11 : 45 am
Minutes:	45 min		
Attendance at Meeting			
Name			
Prof Ricky Chan			
Atal Azadzoi			
Jarred Apps			
Rouhaifa Kakhie			
Shahzad Ahmed			
Shivam Raj			
Discussion			
Topic	Discussed		
Final Review	• Commended effective teamwork and consistent communication.		
Submission Preparation	• Confirmed submission guidelines and naming conventions for digital journal.		
Actioned Items			
Action	Owner / s		
Final edit and proofread report.	Shivam, Rouhaifa		
Ensure naming conventions and file version accuracy.	Jarred		
Upload files for review before submission.	Atal, Shahzad		

Meeting 9			
Project Name:	Real-Time Digital Twin of a Truss Bridge	Date of Meeting:	26 / 09 / 2025
Location:	Microsoft Teams	Time:	10 : 00 am – 10 : 45 am
Minutes:	45 min		
Attendance at Meeting			
Name			
Prof Ricky Chan			
Atal Azadzoi			
Jarred Apps			
Rouhaifa Kakhie			
Shahzad Ahmed			
Shivam Raj			
Discussion			
Topic	Discussed		
Submission and Continuation	• Discussed continuation into Capstone B, focusing on real sensor integration. • Reflected on teamwork efficiency and project outcomes.		
Actioned Items			
Action	Owner / s		
Submit final report and journal on Canvas.	All members		
Save MATLAB data and documentation for Capstone B.	Atal		
Archive communication logs and meeting records.	Shivam		

5. Professional Practice Reflection

Throughout Capstone A, I remained actively engaged in all meetings, deliverables, and peer reviews, taking responsibility for both research direction and technical input. My reflections are outlined below:

1. **Engagement and Leadership**

As Lead Researcher, I guided the development of our research questions and ensured that literature findings were integrated into the design framework. I also supported the team by contributing to MATLAB and software integration tasks, helping link the research outcomes with practical implementation.

2. **Communication and Teamwork**

One of the biggest improvements was our transition from independent writing to fully collaborative editing. Regular peer feedback sessions not only improved the report's consistency but also built team confidence in presenting technical work professionally.

3. **Ethical and Professional Practice**

All research and technical contributions were original, well-referenced, and transparently reported. Simulated data and limitations were clearly acknowledged. I also helped maintain version control and authorship tracking to ensure academic integrity throughout the project.

4. **Technical Development**

Although my primary role was leading research, assisting with MATLAB coding and software planning developed my skills in debugging, structuring data pipelines, and linking theoretical methods with practical tools. This gave me a stronger understanding of how digital-twin systems integrate sensing, modelling, and computation.

5. **Challenges and Growth**

Balancing research leadership with technical contributions and documentation was demanding. Managing both streams required adaptability and time discipline but ultimately strengthened my project-management and prioritisation skills.

6. **Forward Outlook (Capstone B)**

Next semester, I plan to continue leading the research direction while further supporting MATLAB integration. My focus will be on implementing real-time sensor acquisition, refining estimation algorithms such as Kalman filtering, and validating the digital twin against physical bridge behaviour. Capstone A has laid a solid foundation for this next phase.

6. Conclusion

This professional practice and contribution journal highlights my consistent engagement, research leadership, teamwork, and technical growth throughout Capstone A. In my role as Lead Researcher, I guided the development of research questions, integrated literature findings into the project, and ensured alignment between theory and application. Alongside this, I assisted with MATLAB and software integration, supporting the transition from conceptual research into practical digital-twin design.

Collaboration with Professor Ricky Chan and the team strengthened both engineering professionalism and my capacity for independent problem-solving. The foundation built in Capstone A — combining rigorous research, clear documentation, and initial software development — places the project in a strong position to progress into Capstone B, where real-time sensor integration and experimental validation will be achieved.